SmartSteth: A Convolutional Neural Network (CNN) Driven Auscultation Instrument Using Lung Sound Analysis for Early Respiratory Disease Detection in Satellite Clinics

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This study introduces SmartSteth, a portable auscultation device powered by convolutional neural networks (CNN) to support early detection of lower respiratory tract diseases in rural Philippine clinics. The project responds to diagnostic challenges faced by health workers, including limited tools, training gaps, and environmental constraints. Researchers collected and labeled lung sound recordings of normal breathing, asthma, pneumonia, and bronchiolitis. These were converted into Mel-spectrograms and trained using Edge Impulse to build a CNN model for real-time classification. The device integrates an ESP32 microcontroller, MEMS microphone, and a web-based interface that works offline and provides visual feedback. Evaluation showed high sensitivity (≥85 percent), strong F1-score (≥0.83), and reliable performance in noisy settings. Health workers prioritized affordability, portability, and ease of use, which SmartSteth successfully delivered. Compared to existing tools, SmartSteth offers better field readiness, lower cost, and improved accessibility. The study concludes that SmartSteth can strengthen frontline respiratory screening and support equitable healthcare in underserved areas.

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